

Zhiheng (Kevin) Li

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EDUCATION

University of Minnesota, Twin Cities

Minneapolis, MN

Bachelor of Science in **Mathematics** • **Computer Science Minor**

Expected May 2028

- **GPA:** 4.0/4.0 | **Honors:** Dean's List, Fall 2025, Spring 2026 • Tau Sigma Honor Society
- **Relevant Coursework:** Calculus I & II (A), Linear Algebra & Differential Equations (A), Physics for Scientists and Engineers I & II (A), Introduction to Programming Concepts (Python, A), Introduction to Statistical Analysis (A)

RESEARCH EXPERIENCE

SuperCDMS Dark Matter Experiment - School of Physics & Astronomy, University of Minnesota

Minneapolis, MN

Undergraduate Research Assistant & UROP Fellow, Advisor: Prof. Yan Liu

Oct 2025 - Present

- Generated 1×1 and $N \times M$ phonon **pulse templates** for 8 SuperCDMS SNOLAB R4 Ge detectors (up to 12 phonon channels each): applied Butterworth filtering, OFdelay alignment, and 4-exponential fitting; performed PCA on per-event traces for $N \times M$ components; delivered PT/PS1/PS2 composite templates as ROOT output for optimal filter reprocessing (June 2026).
- Built BlueFors-Alert: automated incremental **PostgreSQL sync** (9 tables, every minute) from the BlueFors CS2 Windows PC to a Raspberry Pi via PowerShell scheduled task; cron-based monitor queries 11 cryogenic sensors with Slack alerts and interactive threshold management and acknowledgment commands (June 2026).
- Built a **Python data pipeline** (Oct 2025) in Jupyter Notebook to process LED calibration data from 11 cryogenic Ge detector channels: baseline subtraction, SNR-based glitch detection, peak alignment, and exponential fitting over 5,200+ events (10+ GB raw data); collaborated with postdoc Shubham Pandey on MSI Agate HPC cluster.
- Extended RuOx thermometer **calibration curves** (R31279 & R31839) from factory range to 5 mK–300 K (May 2026); generated validated Lake Shore .340 files using PCHIP + Mott VRH extrapolation for direct controller loading.

SELECTED PROJECTS

College Students Entrepreneurial Project & Entrepreneurship Roadshow

Beijing, China

Team Leader

Oct 2024 - Jun 2025

- Led a student team to design a cost-efficient billiards rest prototype addressing equipment inefficiencies;
- Conducted **market analysis and product testing** to evaluate commercialization potential;
- Invited to present the project at **Tsinghua X-Lab & Kering Entrepreneurship Roadshow**, communicating with a high-profile AI team and presenting to industry leaders.

Low-Carbon Transition Project - China Steel Industry

Beijing, China

Undergraduate Researcher

Oct 2024 - May 2025

- Assisted in research on low-carbon transition pathways for China's steel industry;
- Contributed to **data collection, database construction, and quantitative analysis** for national low-carbon initiatives.

AWARDS & HONORS

- **UROP** (Undergraduate Research Opportunities Program) Fellowship, University of Minnesota, Summer 2026
- **First Prize**, Beijing University Student Innovation & Entrepreneurship Competition (Science & Technology Track)
- **First Prize**, National College E-Commerce "Innovation, Creativity, and Entrepreneurship" Challenge, Campus Level | Team Leader
- **Third Prize**, China International College Students' Innovation Competition, Campus Level (2025) | Team Leader
- **Second Prize**, NECCS National English Competition for College Students

ACTIVITIES & INVOLVEMENT

- **Chinese Student Union**, UMN - Active Member | Oct 2025 - Present
- **Chinese Students & Scholars Association**, UMN - Volunteer Calculus I Tutor | 2025 - Present
- **College Table Tennis Club / Football Club / Guitar Club** - Team Member | 2023 - 2024

PROFESSIONAL SKILLS

- **Techniques:** Cryogenic instrumentation, signal processing, data pipeline development, thermometer calibration curve fitting, real-time PostgreSQL sync, Slack bot development, waveform template construction
- **Tools:** Python, PostgreSQL, Java, NumPy, Pandas, Matplotlib, SciPy, Git/GitHub, Jupyter Notebook, Slurm, MSI HPC (Agate), ROOT, Excel
- **Languages:** Mandarin Chinese (Native), English (Fluent; IELTS 7.0)